Clinical Study Summary Report (CSSR)
Document Status: Final | **Version:** 1.0 | **Date:** 02-Apr-2026

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1. Administrative & Study Identification

Full Study Title: Validating Artificial Intelligence Effectiveness Defined Lung Nodule Malignancy Score in Patients With Pulmonary Nodule **Short Title/Acronym:** CREATE **Protocol Number:** D133FR00178 **NCT Number:** NCT05817110 **Sponsor Name:** AstraZeneca **Investigational Product:** qXR (Artificial Intelligence algorithm by Qure.ai reporting Lung Nodule Malignancy Score - LNMS) **Phase of Development:** Observational **Medical Monitor:** AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To prospectively validate an AI-derived lung nodule malignancy score (qXR-LNMS) against radiologist assessment of CT scans and Lung-RADS in patients with an incidental pulmonary nodule (IPN) on chest radiographs. **Secondary Objectives:** To evaluate the Positive Predictive Value (PPV) and Negative Predictive Value (NPV) of the LNMS by clinico-demographic characteristics, assess concordance with established models (e.g., Mayo Clinic model), and track long-term nodule management and clinical outcomes.

2.2 Background & Rationale While low-dose CT (LDCT) is the gold standard for lung cancer screening, its large-scale adoption in resource-constrained settings is limited by cost and infrastructure. AI-based algorithms (like qXR) applied to routine chest X-rays (CXRs) have demonstrated increased accuracy in predicting lung cancer risk. This study aims to generate real-world evidence validating this AI tool to stratify high- and low-risk patients with IPNs, ultimately optimizing LDCT referrals and early detection, especially for patients falling outside traditional smoking-based screening criteria.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Prospective) **Control Type:** Single-arm (Observational Cohort) **Blinding:** Open-label (Independent radiologist CT review blinded to AI score for validation) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 713 participants **Number of Sites:** Multicentric, multinational (including sites in India, Indonesia, Mexico, Turkey, and Egypt) **Estimated Study Start:** 20-Apr-2023 **Estimated Primary Completion:** 06-Dec-2026

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients aged >35 years. 2. Patients diagnosed with an incidental pulmonary nodule (IPN) on chest X-ray (CXR) ordered by their treating clinician. 3. Nodule detected by the qXR algorithm and confirmed by the site radiologist, with a nodule size ≥ 8 and ≤ 30 mm.

4.2 Exclusion Criteria 1. Patients without a confirmed incidental pulmonary nodule. 2. Inability or refusal to provide written informed consent to participate in the study or to undergo the confirmatory low-dose CT scan. 3. Previously diagnosed with lung cancer or actively undergoing lung cancer treatment.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A (Software as a Medical Device. The AI diagnostic algorithm is applied to a baseline CXR, followed by a confirmatory CT scan). **Route of Administration:** N/A (In silico diagnostic imaging software). **Duration of Treatment:** Baseline imaging assessment (Phase 1) followed by a 24-month observational clinical follow-up period (Phase 2).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for lung nodule management, including follow-up imaging and biopsies as directed by the treating physician. **Prohibited:** N/A (Observational study).

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day 0	Routine CXR performed; qXR algorithm processes the image to detect IPNs.
Baseline (Phase 1)	Enrollment	Informed Consent, AI-based LNMS classification (high/low risk), radiologist confirmation, baseline CT scan requested.
Interim (Phase 2)	Up to Month 24	Clinical follow-up visits per standard of care; not mandatory study-specific visits.
End of Study	Month 24 (post-CT)	Final data collection regarding nodule management, malignancy status, and clinical outcomes.

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Positive predictive value (PPV) and negative predictive value (NPV) of the AI-defined lung nodule malignancy score (qXR-LNMS). (Pre-defined threshold for success: $>20\%$ for PPV and $>70\%$ for NPV). **Measurement Method:** Comparison of the binary qXR-LNMS categorization (high or low risk) with the independent radiologist-based assessment of CT scans (using a 1-5 Likert scale) and Lung-RADS score.

7.2 Secondary & Exploratory Endpoints * Concordance between the AI model and the Mayo Clinic lung cancer risk model. * Subgroup analysis of PPVs and NPVs by clinical and demographic characteristics (e.g., smoking history, age). * Long-term clinical outcomes and diagnostic pathway tracking over 24 months.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: This is a minimal-to-no-risk observational study. AE monitoring is restricted to standard clinical care tracking. **Serious Adverse Events (SAEs):** Reported per standard observational study guidelines. **Data Monitoring Committee (DMC):** Supervised and audited by AstraZeneca and Qure.ai clinical research teams.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled participants with a confirmed IPN who underwent both the AI-analyzed CXR and the subsequent CT scan. **Sample Size Justification:** A target of ~713 participants was determined to provide sufficient statistical power to cross the predefined success thresholds of at least 20% PPV and 70% NPV. **Handling of Missing Data:** Standard imputation methods applicable to observational diagnostic accuracy studies.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Data collection monitoring from initial CXR/qXR assessment (Phase 1) through the 24-month observational period (Phase 2). **Audit Trail:** eCRF tracking of qXR-LNMS scores, local radiologist reports, and independent, blinded radiologist CT scan assessments.

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11. Study Identifiers & Governance

Primary ID: D133FR00178 **Registry Number:** NCT05817110 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** CREATE (Validating Artificial Intelligence Effectiveness Defined Lung Nodule Malignancy Score in Patients With Pulmonary Nodule) **Official Regulatory Title:** A multicentric, multinational, prospective, observational study to validate qXR-Lung nodule malignancy score as a binary categorization of the risk of Lung cancer as high or low among patients with an incidental pulmonary nodule (IPN) on chest radiographs.

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12. Clinical Framework (Lung Nodule Specific)

Primary Condition: Incidental Pulmonary Nodule (IPN) / Lung Malignancy **Diagnosis Threshold:** Nodule size ≥ 8 and ≤ 30 mm on CXR **Medical Methodology:** AI-enabled analysis of chest radiographs (qXR-LNMS) validated against LDCT **Optimization Target:** Early detection and risk-based screening of lung cancer

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13. Study Procedures & Methodology

Management Pathway: Clinical follow-up and LDCT assessment based on standard of care for IPNs **Education Protocol (HCP):** Instruction on qXR AI algorithm reporting, LDCT guidelines, and Lung-RADS scoring **Education Protocol (Patient):** Informed consent for low-dose CT and long-term observational follow-up **Quality Audit Mechanism:** Independent radiologist review of the CT scan compared to the AI-derived LNMS

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14. Observation & Follow-Up Schedule

Total Duration: Approximately 30 months from enrollment (24 months post-CT data collection) **Onsite Visit Cadence:** Baseline CXR and CT Scan visit (Phase 1), followed by routine clinical follow-up (Phase 2) **Remote Monitoring Frequency:** Periodic data collection at End of Phase 1 and throughout Phase 2 **Checklist Review Interval:** Final evaluation at the 24-month follow-up mark

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] |** ___ **|** ***[DD-MMM-YYYY]* | | *Lead Statistician* | *[Name]* |** _____ | [DD-MMM-YYYY] |